

# MICHAEL LEI

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## EDUCATION

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### Duke University

Durham, NC

*B.S. Electrical and Computer Engineering, B.S. Computer Science, GPA: 4.0/4.0*

May 2028

- Relevant Coursework: Software Design, Algorithms, Databases, OS, Networking, Machine Learning, Comp. Architecture

## EXPERIENCE

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### VSee

May 2025 – Present

*AI and Software Engineering Intern*

*Sunnyvale, CA*

- Shipped two production FastAPI services — a clinical de-identification pipeline and an ICD/CPT coding service — that **cut clinical documentation time by 93%**.
- Extended VSee's on-prem LLM platform with MCP integrations (including a custom Elasticsearch analytics MCP server), Langfuse observability, per-user rate limiting, and a HIPAA-safe Vertex AI path, **increasing usage by 17%**.
- Built an LLM code-review service triggered by PR webhooks, reviewing **100+ production PRs across 3 repos**.
- Designed a benchmarking harness evaluating both self-hosted and cloud speech-to-text providers over WebSocket streaming, scoring with a custom medical-WER metric backed by a spaCy NLP pipeline.

### Forewarned

May 2026 - Present

*Software Engineer*

*Durham, NC*

- Built cloud microservices unifying real-time block telemetry from Palo Alto, AWS, Azure, and edge routers into a single schema, with an AI layer ranking persistent attackers and surfacing coordinated campaigns no single device could see.
- Correlated perimeter block events with **honeypot signals** to flag repeat offenders across the platform's 80+ partner network.

### Chilkoti Group

Summer 2025

*Machine Learning Engineer*

*Remote*

- Built an OpenCV-based preprocessing and parameter auto-tuning pipeline for assay slide annotation as part of the workflow producing ground-truth training data for cancer-screening assay ML classifiers.
- Designed a **coarse-to-fine grid search** algorithm over Hough-circle parameters, replacing per-slide manual slider-tuning.

### Duke University

Fall 2024

*Software Engineer*

*Durham, NC*

- Engineered C++ firmware for an ESP32 flagpole controller built for Duke grounds staff, implementing an **MQTT control plane** and **ACS712-based motor stall detection** to automate a manual flag-changing process.
- Designed the MQTT topic schema so that adding the Nth flagpole requires **zero frontend changes**, devices auto-announce via retained topics and a broadcast topic fans one click to every pole.

## PROJECTS

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### DukeDine | *Python, React, PostgreSQL, AWS* | [Demo](#)

- Built and shipped **DukeDine**, a full-stack data platform for Duke students aggregating **structured performance metrics and unstructured dining data** into a searchable Postgres backend.
- Designed an **8-agent Claude system on AWS Bedrock** with forced-clarification tool-use and a **pgvector RAG layer** that regex-verifies every cited date and quote against retrieved data before a response ships.

### Bet308 | *Java, JavaFX* | [GitHub](#)

- Built **Bet308**, a **data-driven card-game engine** with an **LLM-assisted authoring environment** that turns natural-language prompts into playable game JSON, shipping seven games on a single interpreter plus multiplayer.
- Worked in an 8-person team through **6 weeks of agile development**, running one-week shippable sprints.

### Wildfire AI Drone | *Python, Swift, DJI Mobile SDK*

- Built a **wildfire detection system** that ran a trained detector **on-device against a drone's live video feed**, integrating the drone SDK into a per-frame inference path.
- Named **U.S. Division Semi-Finalist** at the 2022 China-US Young Maker Competition.

## TECHNICAL SKILLS

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**Languages:** Python, TypeScript / JavaScript, C++, Java, SQL

**Web & Backend:** FastAPI, React, Node.js, HTML/CSS, REST APIs, PostgreSQL

**AI-Assistant Development:** Cursor, Claude Code, LLM Prompting, Code Review, Agentic Workflows

**Systems & Tools:** Linux/Unix, Docker, Git, AWS, GitHub Actions, Bitbucket Pipelines, Metrics, Debugging